PROGRAMMING METHODS - LABORATORY 11

Program 01

Create a class cTablica containing the following array operations:

- bubble sort - shaker sort variant;
- quick sort using partition algorithms: (1) Lomuto and (2) Hoare;
- heap sort.

In the sorting methods a-c, insert instructions that count the number of comparisons and the number of swaps.

Create a class cSortTablicy that allows:

- entering a specified number of elements from the keyboard;
- randomizing a selected number of elements;
- selecting and illustrating how the sorting methods work for arrays of different sizes (100, 1000, 10,000,000) and values: random, sorted ascending, reverse sorted, and initially sorted with 10% of elements in wrong positions;
- for every data set and every sorting method, printing the number of comparisons and swaps.

Use exceptions to handle errors.

Required result file

Create a text file presenting the obtained results in table form:

Sorting method name	Array length	Array type	Number of comparisons	Number of swaps
		array filled with random numbers		
		array filled with ascending sorted numbers		
		array filled with descending sorted numbers		
		partially ordered array (10% of elements are in wrong positions)		

What conclusions do you reach? Put your observations in the output text file, below the generated table.

Submission rules

The .cpp file(s) should be named:

```
Surname_Name_Program_01.cpp
```

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_11.

Submit the folder to: Programs - laboratory 11 - Tuesday.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 11 on delta.pk.edu.pl.